Timeline Check-In

Table of contents

Set Up

Packages

```
# load in tidyverse package
library(tidyverse)
# load in here package
library(here)
# load in janitor package
library(janitor)
# load in snakecase package
library(snakecase)
# load in scales package
library(scales)
# load in readxl package
library(readxl)
# load in naniar package
library(naniar)
# load in patchwork package
library(patchwork)
# load in dplyr package
library(dplyr)
# load in broom package
library(broom)
# load in corrplot package
library(corrplot)
```

```r
# load in ggrepel package
library(ggrepel)
```

Data

```r
# read in taxonomic info data
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# read in aquatic ivert. data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# read in site data
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# read in water quality data
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning `taxon_list`

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning `water_quality`

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning `aq_ins`

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
```

```r
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

Focus Taxa (`focus_taxa`)

```r
# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                "Hemiptera Corixidae Boatman", "Nematode")
```

Creating DO_taxa, total_abundance, and DO bins

```
# create a new objecy aq_ins_clean_bins that adds DO bins to aq_ins_clean
aq_ins_clean_bins <- aq_ins_clean |>
  # create a new bins column based on site_full values
  mutate(bins = case_when(site_full == "Phelps Road" ~ "low",
                           site_full == "North of Phelps Bridge" ~ "low",
                           site_full == "South of Venoco Bridge" ~ "moderate",
                           site_full == "South of Phelps Bridge" ~ "moderate",
                           site_full =="Devereux Creek" ~ "moderate",
                           site_full =="North of Venoco Bridge" ~ "high",
                           site_full == "Main Channel" ~ "high")) |>
  # replace underscores in taxon with spaces & convertto title case
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # keep only taxa included in the focus_taxa object
  filter(taxon %in% focus_taxa) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert bins into an ordered factor
  mutate(bins = factor(bins, levels = c("low", "moderate", "high")))

# create object DO_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)

# create objetc total_abundance from DO_taxa
total_abundance <- DO_taxa |>
  # keep only taxa included in the focus_taxa object
  filter(taxon %in% focus_taxa) |>
  # group observations by date, site, and median DO
  group_by(date, site, med_do) |>
```

```
  # sum average abundance per liter across taxa & ignore missing values
  summarize(total_abundance = sum(ave_lit, na.rm = TRUE),
            # ungroup
            .groups = "drop") |>
  # create a log10-transformed total abundance column called log_total_abundance
  mutate(log_total_abundance = log10(total_abundance + 1))

# updates the total_abundance object
total_abundance <- total_abundance |>
  # create a new variable do_category based on DO quantiles
  mutate(
    # classify each observation into low, medium, or high dissolved oxygen
    do_category = case_when(
      # assign Low DO to values at or below the 25th percentile of median DO
      med_do <= quantile(med_do, 0.25, na.rm = TRUE) ~ "Low DO",
      # assign Medium DO to values between the 25th percentile - 75th percentile
      med_do <= quantile(med_do, 0.75, na.rm = TRUE) ~ "Medium DO",
      # assign High DO to all remaining values above the 75th percentile
      TRUE ~ "High DO"))

# edit the column do_category into an ordered factor
total_abundance$do_category <- factor(total_abundance$do_category,
                                      # set the category order
                                      levels = c("Low DO", "Medium DO", "High DO"))
```

Ideas for analysis and background

Ideas for analysis:

Our first big idea is looking at total invertebrate abundance compared to median dissolved oxygen levels. We ran a correlation test and found that there was a statistically significant negative linear relationship between median dissolved oxygen and log-transformed total invertebrate abundance, suggesting that abundance decreased as DO increased.

We also explored how dissolved oxygen levels differ between sites, and then categorized sites into low, moderate, and high DO bins. We found that total log-transformed invertebrate abundance did not differ significantly across low, moderate, and high DO bins. However, Ostracod abundance differed significantly across DO bins, indicating that Ostracods may respond differently to low, moderate, and high DO conditions. After Bonferroni correction, the moderate vs. high DO comparison was close to significant, suggesting a possible difference in Ostracod abundance between these bins but not strong enough to conclude significance at $= 0.05$.

Another way we compared taxa in bins was by running a Chi-square test on taxon composition within each DO bin. We found that taxon composition differed significantly across DO bins, suggesting that the invertebrate community structure changes depending on dissolved oxygen category. Next, we'd like to perform post-hoc tests to see which groups differ from one another.

Background:

North Campus Open Space is a restored wetland, which makes it a great area to explore data about the effects of restoration. One metric of restoration progress is water quality, and dissolved oxygen concentration is a key water-quality indicator. We are interested in the relationship between invertebrate abundance and dissolved oxygen, which ultimately ties back to the health of this wetland. This is a less traditional method of measuring ecosystem health, but can become more widely used with increasing data showing these relationships. This study applies to all wetland ecosystems and aquatic environments more broadly. In addition to being a metric of ecosystem health, invertebrate abundance compared to dissolved oxygen can tell us about the conditions that these invertebrates exist in.

Citations:

Poster: Algae, Macroinvertebrate and Water Quality Relationships at North Campus Open Space - link

Background Articles

Connolly, N. M., Crossland, M. R., & Pearson, R. G. (2004). *Effect of low dissolved oxygen on survival, emergence, and drift of tropical stream macroinvertebrates.* BioOne Complete (BioOne). https://doi.org/10.1899/0887-3593(2004)023%3C0251:eoldoo%3E2.0.co;2

Gunsch, D. E. (2020, November 5). *Community Composition of Aquatic Invertebrates along Dissolved Oxygen Gradients in Lake Huron Coastal Wetland Wet Meadows.* Handle.net. https://hdl.handle.net/20.500.14776/9073

Lucchesi, D. O., Chipps, S. R., & Schumann, D. A. (2022). *Effects of seasonal hypoxia on macroinvertebrate communities in a small reservoir.* Lakes & Reservoirs: Science, Policy and Management for Sustainable Use, 27(1). https://doi.org/10.1111/lre.12395

Suren, A. M., Lambert, P., & Sorrell, B. K. (2010). *The Impact of Hydrological Restoration on Benthic Aquatic Invertebrate Communities in a New Zealand Wetland.* Restoration Ecology, 19(6), 747–757. https://doi.org/10.1111/j.1526-100x.2010.00723.x

Spearman Correlation Analysis for Total Abundance and DO levels

```
# run a Spearman correlation test between log total abundance & median DO
cor.test(
  # use log-transformed total abundance as the first variable
  total_abundance$log_total_abundance,
  # use median DO as the second variable
  total_abundance$med_do,
  # specify Spearman method
  method = "spearman",
  # use approximate p-values
  exact = FALSE)
```

```
    Spearman's rank correlation rho

data:  total_abundance$log_total_abundance and total_abundance$med_do
S = 8526.3, p-value = 0.08182
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.3027201
```

Total log-transformed invertebrate abundance had a **weak negative** relationship with dissolved oxygen, but this relationship was not statistically significant using Spearman correlation.

Correlation analysis for Each Focus Taxa and DO levels

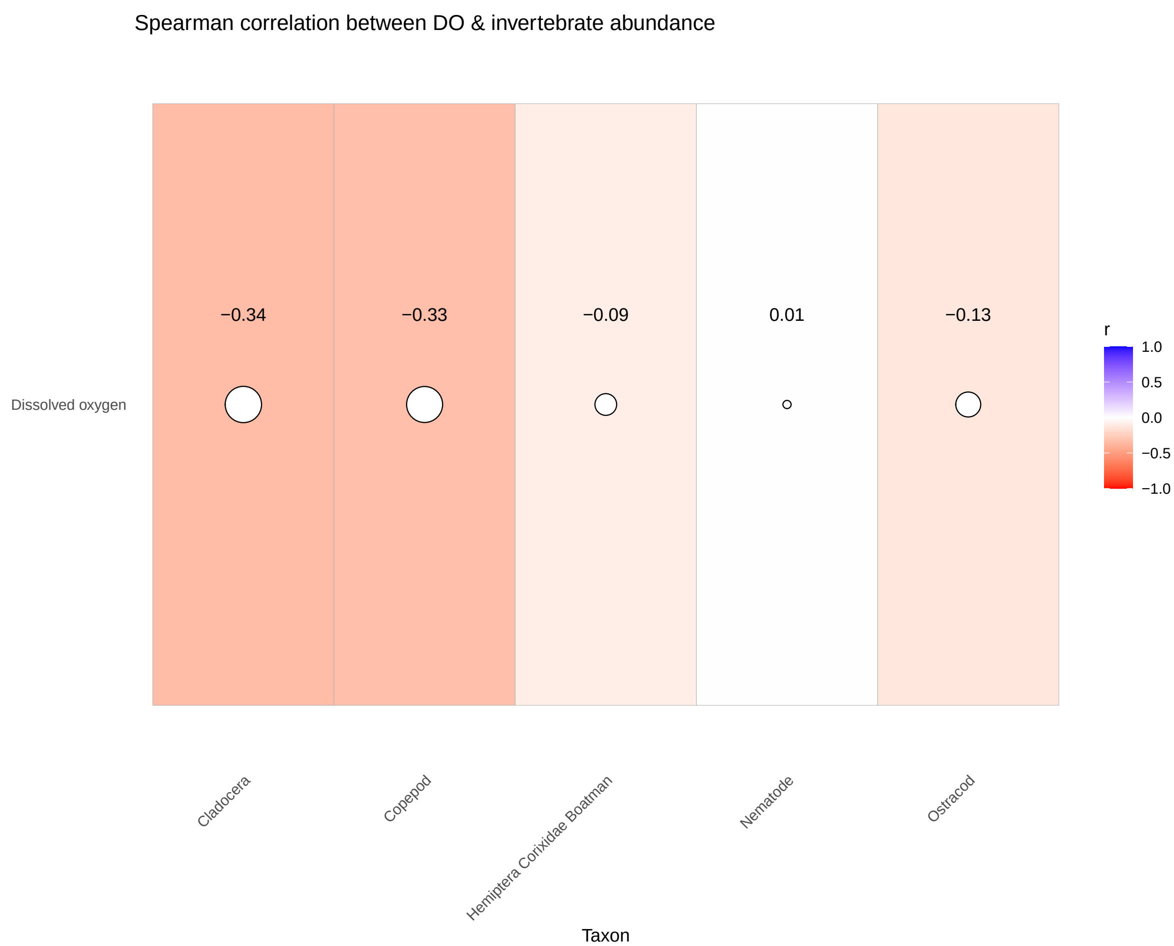
```
# create wide dataset (one column per taxon)
taxa_wide <- DO_taxa |>
  # keep only the columns needed
  select(date, site, taxon, log_ave_lit, med_do) |>
  # convert the dataset from long format to wide format
  pivot_wider(names_from = taxon, # create one new column for each taxon
              # fill each taxon column with log-transformed abundance values
              values_from = log_ave_lit)

# calculate correlation of each taxon with dissolved oxygen
```

```r
# calculate Spearman correlation of each taxon with dissolved oxygen
cor_do <- taxa_wide |>
  # remove date and site columns so only DO and taxon abundance columns remain
  select(-date, -site) |>
  # calculate correlations for each taxon column
  summarize(across(-med_do, ~ cor(.x,
                                  # correlate each column w/ median DO
                                  med_do,
                                  # specify spearman method
                                  method = "spearman",
                                  # only use rows w/ complete observations
                                  use = "complete.obs"))) |>
  # convert the correlation results from wide format to long format
  pivot_longer(cols = everything(),
               # store taxon names in a column called taxon
               names_to = "taxon",
               # store correlation values in a column called correlation
               values_to = "correlation")

ggplot(cor_do, # create a new plot from cor_do
       aes(x = taxon, # x-axis is taxon
           y = "Dissolved oxygen", # titles y-axis
           fill = correlation)) + # fill color by correlation
  # add tiles where fill represents the strength + direction of correlation
  geom_tile(color = "grey70") +
  # add circles sized by the absolute value of the correlation
  geom_point(aes(size = abs(correlation)), shape = 21, fill = "white") +
  # add labels above circles
  geom_text(aes(label = round(correlation, 2)),
            # move the text labels slightly above the circles
            nudge_y = 0.15, size = 4) +
  # create a color scale for negative, neutral, and positive correlations
  scale_fill_gradient2(
    # use red for negative correlations
    low = "red",
    # use white for correlations near zero
    mid = "white",
    # use blue for positive correlations
    high = "blue",
    # sets midpoint to 0
    midpoint = 0,
    # set the color scale limits from -1 to 1
```

```
  limits = c(-1, 1),
  # labels legend 'r'
  name = "r") +
# set the range of circle sizes + remove the size legend
scale_size(range = c(2, 10), guide = "none") +
labs(x = "Taxon", # labels x-axis
     y = NULL, # remove the y-axis label
     # creates descriptive title
     title = "Spearman correlation between DO & invertebrate abundance") +
# usese non-default theme
theme_minimal() +
theme(panel.grid = element_blank(), # remove panel grid lines
      # rotate x-axis labels to improve readability
      axis.text.x = element_text(angle = 45, hjust = 1))
```



Most focus taxa showed **weak negative** correlations with dissolved oxygen, with Cladocera and Copepod showing the strongest negative relationships and Nematode showing almost no relationship.

Pearson correlation for continuous DO and total community abundance

```
# run a Pearson correlation test between log total abundance + median DO
cor.test(
  total_abundance$log_total_abundance, # log total abundance is first variable
  total_abundance$med_do, # use median DO as the second variable
         method = "pearson") # specify pearson method
```

```

	Pearson's product-moment correlation

data:  total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.63289820 -0.04217141
sample estimates:
       cor 
-0.3749894 
```

```
# r = -0.3749894, p = 0.02887
# there is a significant negative correlation between abundance and continous DO
```

There was a **statistically significant negative linear relationship** between median dissolved oxygen and log-transformed total invertebrate abundance, suggesting that abundance decreased as DO increased.

Kruskal-Wallis test comparing abundance across DO categories

```
# run a Kruskal-Wallis test comparing log total abundance across DO categories
kruskal.test(log_total_abundance ~ do_category,
             # use the total_abundance dataset
             data = total_abundance)
```

```
    Kruskal-Wallis rank sum test

data:  log_total_abundance by do_category
Kruskal-Wallis chi-squared = 2.0001, df = 2, p-value = 0.3679
```

```
# Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
# there is not a significant difference between abundance and DO categories
```

Total log-transformed invertebrate abundance **did not differ significantly** across low, moderate, and high DO categories.

Chi-Square

```
# create abundance table for chi-square test
chi_square_table <- aq_ins_clean_bins |>
  # keep only rows with the focus taxa
  filter(taxon %in% focus_taxa) |>
  # group data by DO bin + taxon
  group_by(bins, taxon) |>
  summarise(
    # sum average abundance per liter + ignore missing values
    total_abundance = sum(ave_lit, na.rm = TRUE),
    # ungroup
    .groups = "drop") |>
  # convert the table from long format to wide format
  pivot_wider(
    # create one column for each taxon
    names_from = taxon,
    # fill taxon columns with total abundance values
    values_from = total_abundance,
    # fill missing values with 0
    values_fill = 0) |>
  # convert the bins column into row names
  column_to_rownames("bins") |>
  # convert the table into a matrix
  as.matrix()

# show table
chi_square_table
```

```
         Cladocera   Copepod Hemiptera Corixidae Boatman  Nematode    Ostracod
low       980.2667  115.8667                   0.2666667 0.1333333 1202.933333
moderate  510.6667 1170.4000                  83.8666667 0.2666667 1057.866667
high      264.2667 1453.4667                   8.8000000 0.0000000    7.866667
```

```
# chi-square test for differences in taxon composition across DO bins
chi_square_test <- chisq.test(chi_square_table)

# show chi-square result
chi_square_test
```

```

    Pearson's Chi-squared test

data:  chi_square_table
X-squared = 2859.2, df = 8, p-value < 2.2e-16
```

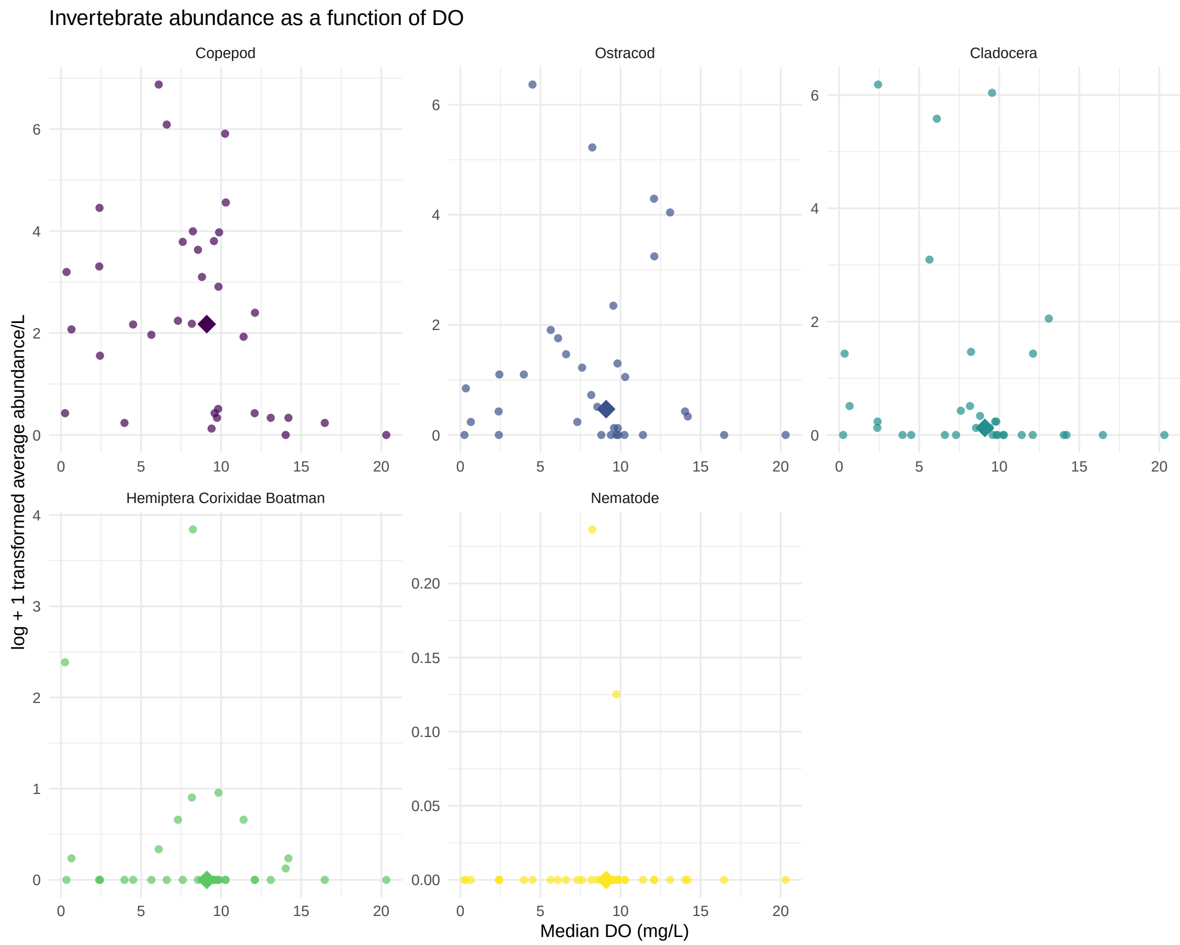
Taxon composition **differed significantly across DO bins**, suggesting that the invertebrate community structure changes depending on dissolved oxygen category.

Visualizations directly relevant to answering questions

Directly Relevant Figure 1: DO levels vs invert. abundance for each focus taxa

```
# creates object DO_taxa_medians from DO_taxa
DO_taxa_medians <- DO_taxa |>
  # groups data by taxon so each species gets one median point
  group_by(taxon) |>
  # calculates median DO and median log abundance for each taxon
  summarise(
    # calculates the median DO value for each taxon
    med_do_point = median(med_do, na.rm = TRUE),
    # calculates the median log-transformed average abundance for each taxon
    med_log_ave_lit = median(log_ave_lit, na.rm = TRUE),
    # ungroup
    .groups = "drop")
```

```r
DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
                       # uses DO_taxa data frame
                       mapping = aes(x = med_do, # x-axis: med_do
                                     y = log_ave_lit, # y-axis: log_ave_lit
                                     color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
             size = 2, # size is 2
             alpha = 0.7) + # opacity is 0.7
  # second layer: larger point shows the median location for each taxon
  geom_point(data = DO_taxa_medians,
             mapping = aes(x = med_do_point, # x value is median DO
                           # y value is median log-transformed abundance
                           y = med_log_ave_lit),
             shape = 18, # makes shape of points diamond
             size = 5) + # size of points is 5
  # separate panels for each taxon with free x- and y-axis scales
  facet_wrap(~ taxon, scales = "free") +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "Median DO (mg/L)", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance as a function of DO") +
  # uses non-default theme
  theme_minimal() +
  # remove redundant legend
  theme(legend.position = "none")
# shows plot
DO_taxa_plot
```



Description of visual components

This figure shows the relationship between Median DO (mg/L) and log + 1 transformed average invertebrate abundance/L for our five focus taxa. Each panel represents a different taxon. Each small point represents one observation of that taxon, with median dissolved oxygen shown on the x-axis in mg/L and log + 1 transformed average abundance per liter shown on the y-axis. The different colors distinguish the taxa, and the larger diamond in each panel represents the overall average/summary point for that taxon across observations.

Main message / takeaway

Overall, there does not appear to be a strong or consistent relationship between median dissolved oxygen and invertebrate abundance across all taxa. Some taxa, especially Copepods, Ostracods, and Cladocerans, show higher abundances at low to moderate dissolved oxygen levels, but the pattern is variable and differs by taxon.

Connection to the main question

This figure helps evaluate whether dissolved oxygen is related to invertebrate abundance in the study system. The variation among taxa suggests that dissolved oxygen may influence some invertebrate groups differently, but DO alone does not clearly explain abundance patterns across the full invertebrate community.

Directly Relevant Figure 2: DO levels vs total invert. abundance

```r
# fit linear model for relationship between log abundance and median DO
total_abundance_lm <- lm(log_total_abundance ~ med_do,
                         # use total_abundance as df
                         data = total_abundance)

# create new data frame total_abundance_predictions for predicted values
total_abundance_predictions <- data.frame(
  # create a sequence of DO values from the minimum to maximum observed DO
  med_do = seq(min(total_abundance$med_do, na.rm = TRUE), # max value is max DO
               max(total_abundance$med_do, na.rm = TRUE), # min value is min DO
               # creates 100 evenly spaced values for smooth predictions
               length.out = 100))

# generate predicted values and confidence intervals
total_abundance_predictions <- total_abundance_predictions |>
  bind_cols( # attach model predictions + confidence interval values to df
    # predict log total abundance using the fitted linear model
    predict(total_abundance_lm,
            # use the total_abundance_predictions df as data for prediction
            newdata = total_abundance_predictions,
            # create confidence intervals around the fitted values
            interval = "confidence") |>
      # convert prediction output into a data frame
      as.data.frame())

# creat plot from total_abundance df
ggplot(total_abundance,
       aes(x = med_do, # x-axis is total_abundance
           y = log_total_abundance)) + # y-axis is log_total_abundance
  # layer 1: create points
  geom_point(color = "magenta3", # color points
             size = 3, # size of points is 3
             alpha = 0.7) + # opacity of points is 0.7
  # layer 2: creates ribbon confidence interval around model predictions
```